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Structuring architecture with salvaged timber: Exploring an interlocking modular system and beyond

G. Ruan
Department of Civil Engineering, Aalto University, Espoo, Finland

G.H. Filz
Faculty of Architecture, Universität Innsbruck, Innsbruck, Austria

G. Fink
Department of Civil Engineering, Aalto University, Espoo, Finland

ABSTRACT: This paper explores an interlocking modular system constructed from salvaged timber through an integrated structural and architectural design process. Starting from a simple reciprocal frame concept, two design cases were developed from different types of deriving logic. Physical prototypes were constructed at full scale using locally salvaged timber in both cases. An experimental loading test was performed for one prototype under specific loading and boundary conditions according to its potential structural application. The test results demonstrate the structural potentials and limits of the prototype. Through rethinking the design, making and testing of the presented cases, the relationship of structure and architecture is extensively discussed, where structuring architecture is proposed as an alternative approach to pursuing sustainability.

Keywords: structuring architecture, salvaged timber, wood interlocking system, modular fabrication, sustainability

1 INTRODUCTION

The current environmental crisis demands a shift in how we build, moving towards a circular design-build approach. In circular design, no material is considered waste; instead, used materials are reused as raw materials for new products. A key challenge for this approach lies in the variability of reused or salvaged materials, which often have non-standard dimensions and varied qualities. This brings difficulties when using traditional design approach, where material-sizing is usually finalised by the end of design process. To address it, design approaches based on material availability were proposed, allowing the characteristics of salvaged materials to inform design, as showcased in some recent projects (e.g., Browne et al. 2022, Ruan et al. 2023). In fact, this is not a new concept, as similar practices are found in vernacular timber architecture, where timber used for construction is largely determined by the availability of local forests (Zwerger 2023).

Nowadays, timber is widely used in construction, from massive load-bearing components to temporary supports. Reusing construction timber, particularly as structural elements, could offer ecological, social, and economic benefits. While timber are possibly reused for structural purposes (Falk et al. 2008), the primary challenge for design lies in the material's irregularity. Although timber products generally have standardised cross-sections, the varying and short lengths of salvaged timber must be carefully managed during the design process. In this context, traditional wood interlocking techniques may provide valuable insights. Wood interlock is created by shaping two or more wooden parts to form specific geometric features that mechanically tighten the joints. It offers two significant benefits for reusing short timber elements with varying lengths: (i)

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timber elements can be joined without additional materials to create load-bearing structures longer than the individual elements, and (ii) some wood interlocks, such as mortise-tenon joint, allow for over-lengths, enabling the use of timber in varying lengths without cutting.

Motivated by the above-mentioned benefits offered by wood interlock, a specific interlocking system—reciprocal frame (RF)—has been explored in the authors' five-year collaborative research project on the topic of *Timber-only Structures & Architecture* (Ruan 2024), where *form*, as a media to unite *structure* and *architecture*, has been specifically discussed. By specifically using salvaged timber as the only *material* for main structures, the project demonstrated how a limited material input can be effectively utilised, i.e., reducing material processing and waste, and offering sufficient load-bearing capacity, to construct load-bearing structures with selected forms (Fink et al. 2019, Ruan et al. 2021 & 2022a).

As a continuous investigation on wood interlocking system using salvaged timber, this paper explores a specific load-bearing modular system based on a simple RF concept. Two design cases developed from different types of deriving logic are presented, showcasing their assembly and disassembly approaches. Physical prototypes were constructed at 1:1 scale using locally salvaged timber. Furthermore, an experimental loading test was performed in the second case under specific loading and boundary conditions. Through rethinking design, making and testing processes of the cases, the relationship of structure and architecture is extensively discussed, where *structuring architecture* is proposed as an alternative approach to pursuing sustainability.

2 DESIGN CONCEPT

The proposed design concept was inspired by two types of interlocking structures. One is *slide-in RF system* that forms slide-in connections through a specific layered arrangement of its individual elements, i.e., leaving an opening in the middle layer(s) so that other elements can be inserted to form a closed reciprocal system (see Figure 1). The other one is *Dougong bracket system*, which is commonly used for connecting columns and upper roof structures in ancient Asian architecture (see Figure 2). The two structures both have a layered and modular arrangement, which allows the use of short elements with varying lengths. An integration of the features of these two structures was therefore explored using salvaged timber, aiming at a minimal processing design concept.

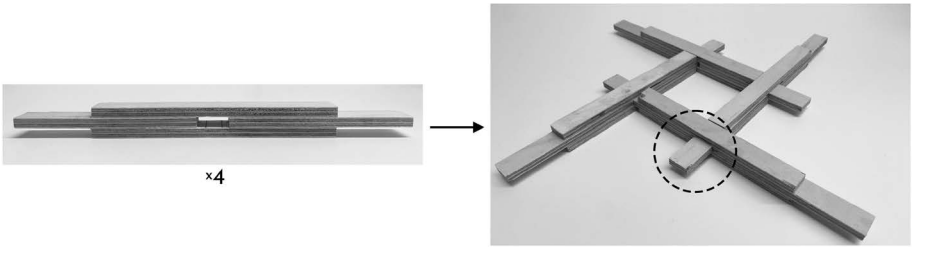


Figure 1. An example of four-element slide-in RF system assembled from individual element (left) through slide-in connection (right) (Ruan et al. 2022a).



Figure 2. A typical Dougong bracket in an ancient Chinese temple.

Dougong bracket system is typically arranged as an inverted triangular shape. This arrangement facilitates load transfer from roof down to columns, avoiding load concentrations. Building on this concept, a system was developed from a basic four-element slide-in RF system by increasing number of layers and gradually scaling down element size from top to bottom. It can be derived with two types of repeating logic: (i) *reverse repeat* and (ii) *step repeat*. To describe the repeating logic and the developed structures, terms such as *layer*, *element*, *unit* and *system* are defined. *Layers* are salvaged timber boards that are connected with fasteners (wooden or steel nails) to form an *element*, and they should be arranged in a way that at least one opening is left to form slide-in connection, as seen in Figure 1; each *element* has three or more *layers*; a basic rectangular *unit* consists of four *elements*; and a *system* has one or more *units* that are connected together.

Both starting from a basic four-element unit (in black colour in Figure 3), the two types of deriving logic can result in different patterns. *Reverse repeat* represents a repeating logic that forms a reverse pattern when a unit scales up or down, as seen in Figure 3 (left) where the unit in dark grey colour has a reverse pattern compared to the larger unit (in black colour) and the smaller unit (in light grey colour). *Step repeat* represents a repeating logic that forms a similar pattern when a unit scales up or down, as seen in Figure 3 (right) where the basic unit (in black colour) is gradually scaled down four times to form continuous five-layer steps (colour changes from dark to light). As demonstrations, two scaled physical models were made. For each demonstrator, a specific sequence needs to be followed for assembly, as highlighted in Figure 3. For more detailed information of the assembly of slide-in RF system, Ruan et al. (2022a & 2024) are referred to.

3 DESIGN CASES

Two design cases were developed according to the two types of deriving logic and built at 1:1 scale using locally salvaged timber. The first case demonstrates *reverse repeat* in a canopy structure, while the second case demonstrates *step repeat* in a pyramid-like structural component.

3.1 *Case-I: A canopy structure*

Case-I is a canopy structure that has two main parts: a superstructure and four supporting columns (see Figure 4). The superstructure consists of four substructures, each containing three

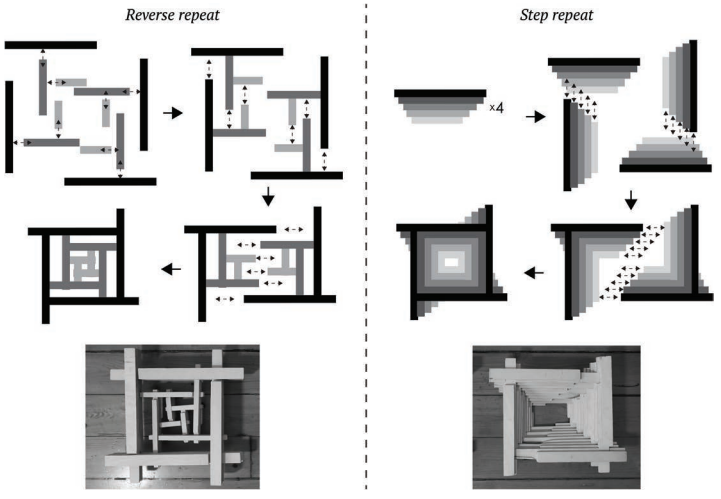


Figure 3. *Reverse repeat* and *step repeat*: assembly sequence and physical model to demonstrate the two types of deriving logic.

different unit sizes and allowing a column inserted into the inner hole of the smallest unit. Each substructure applies the concept of *reverse repeat*, resulting in a similar pattern with varying proportions. A total of 48 elements were designed and prefabricated from reused timber boards with a specific cross-section (40 mm $\times$120 mm) but varying lengths using a zero-waste principle, i.e., fully utilise the original timber boards through efficient cutting. The flexibility of the slide-in system, which allows for over-length connections, made this possible. The four columns were constructed on-site first. Since only manual labours were available for the building process, assembling the entire superstructure before placing on the columns was impractical due to its approximately 500 kg of self-weight. Instead, the elements were assembled as individual substructures (following the process in Figure 3 (left)) and installed on the columns. To achieve the final configuration, in-plane rotational sliding between the substructures was necessary, which was accomplished by rotating the outer units through pre-reserved gaps in the outer elements. The detailed assembly process is illustrated in Figure 4. The canopy was fabricated and constructed at the Faculty of Architecture, Universität Innsbruck. The finalised structure, the patterns, connection details, and resulting shadows are shown in Figure 5.

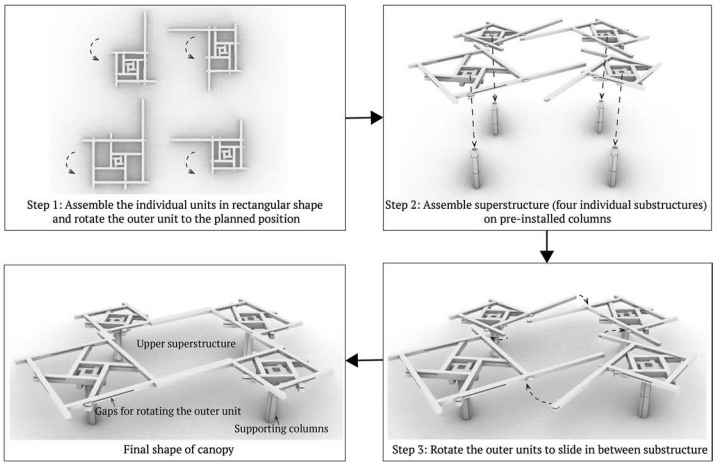


Figure 4. Case-I: assembly process from individual separated units to final shape through slide-in.

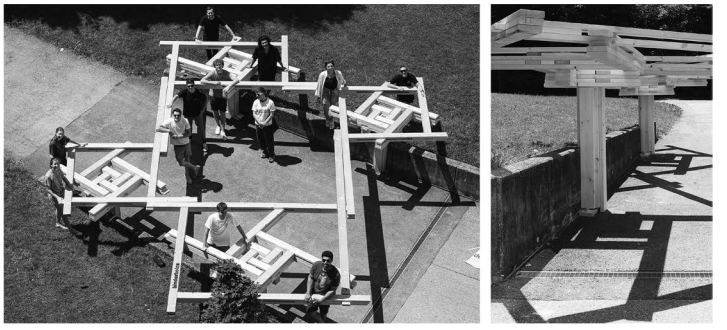


Figure 5. Case-I constructed on site: finalised structure (left) and the resulting shadows (right).

3.2 *Case-II: A pyramid-like structural component*

Case-II is a demonstration of *step repeat* in full scale. A prototype that has a pyramid-like geometry was designed with a constantly reduced unit size from top to bottom. A step length of 30 mm was adopted between each two vertical neighbouring layers to ensure enough overlapping area for placing fasteners. The materials used for constructing the prototype were salvaged

ungraded timber from a local construction site with a thickness of t = 40 mm and a width of w = 100 mm, and varying and short lengths (l < 1500 mm). Wooden nails with 75 mm-length and 4.7 mm-diameter were used for connecting each two layers perpendicular to the grain. The prototype was designed in a way that only the minimal length for each element was given, as seen in Step 1 in Figure 6. The number of layers was fully determined by the amount of materials in storage. As a result, 15 layers were obtained with hole sizes (the size of inner square of one layer) changing from d^{min} = 200 mm at the bottom to d^{max} = 1100 mm at the top (detailed definition is also shown in Figure 9). Through selecting suitable timber boards for each layer and constructing from bottom to top, the prototype was realised without material waste. The prototype was constructed in a layer-by-layer approach (Step 2 in Figure 6) rather than the element-by-element approach illustrated in Figure 3. The reason is that the difficulty of assembly will increase when there are excessive layers, due to the increased frictions and potential misfits caused by the slide-in connections. However, this still provides a possibility to disassemble the entire structure as four large elements (see Step 4 in Figure 6). The finalised prototype and the highlight of connecting details are shown in Figure 7.

As a potential application in buildings, the prototype can be used similarly with the Dougong bracket system, i.e., connecting roof or floor to columns. Following this idea, Figure 8 illustrates a typical loading condition, where the prototype, as a structural component, is exposed to compression. To investigate its structural behaviour, an experimental loading test was carried out at the Department of Civil Engineering, Aalto University. The full-scale

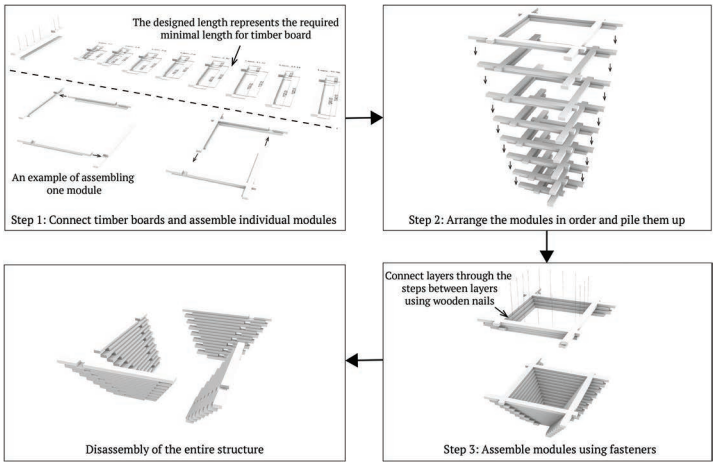


Figure 6. Case-II: assembly process from individual layers with different lengths to final shape.



Figure 7. Case-II in an exhibition: the atmosphere in designated space (left) and details (right).

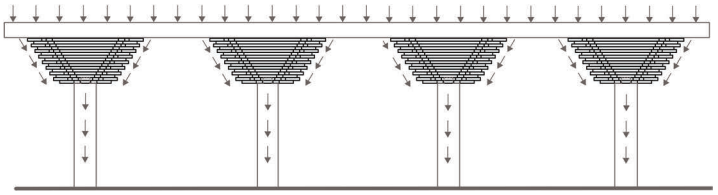


Figure 8. Load transfer in a roof-column system using the proposed structural component.

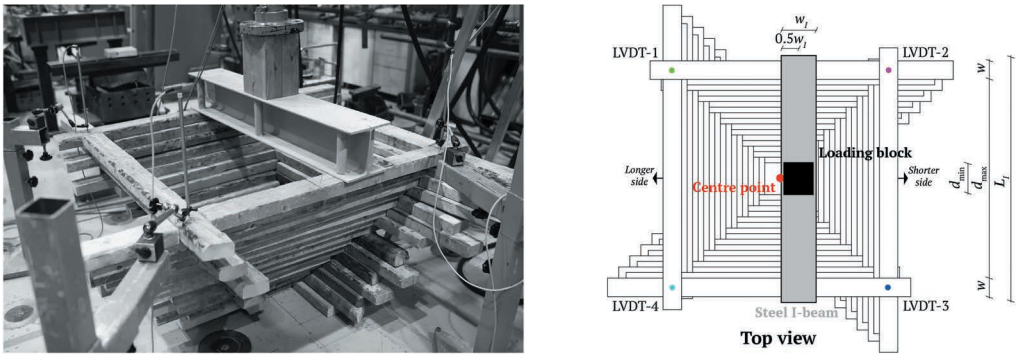


Figure 9. Setup of the eccentric compressive loading test.

prototype was placed on floor (assuming its connection to column rigid enough to carry loads) and loaded vertically by a hydraulic loading device with a capacity of 200 kN. Considering the loading condition shown in Figure 8 that results in eccentric loads under evenly distributed loads on the roof, an eccentric loading setup was applied for the loading test. The load was applied on the middle of a steel I-beam (length: L_I = 1400 mm; flange width: w_I = 200 mm), which was placed 100 mm offset from the centre of the prototype. Detailed information of the test setup is shown in Figure 9. To measure vertical displacement of the prototype, a linear variable differential transformer (LVDT) was placed at each corner of the prototype. The prototype was loaded until failure.

The maximum load achieved in the test was 148.8 kN. Compressive deformation perpendicular to the grain was developed during the entire loading process due to the high stress concentration near the steel beam (Figure 10). The final failure mode was tensile or combined tension-shear failure of wooden nails between the second and third top layers near the corner where LVDT-4 stands (Figure 11). The reason is that after the initial gaps between layers (due to manufacturing) were closed under the compressive loads, the shorter side (see its definition in Figure 9) was still subjected to compression, while the longer side (see also Figure 9) started to carry tension (possibly combined with shear, as nails were not perfectly arranged perpendicular to the grain), leading to the final failure of nails. This phenomena can be also explained by the LVDT's measurements (see Figure 12): the measurements of LVDT-2 and -3 located at the shorter side continuously increased until failure,



Figure 10. Compressive deformation perpendicular to the grain due to high stress concentration.

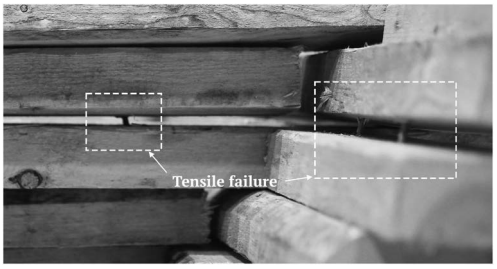


Figure 11. Failure mode: tensile failure of wooden nails (possibly combined with shear).

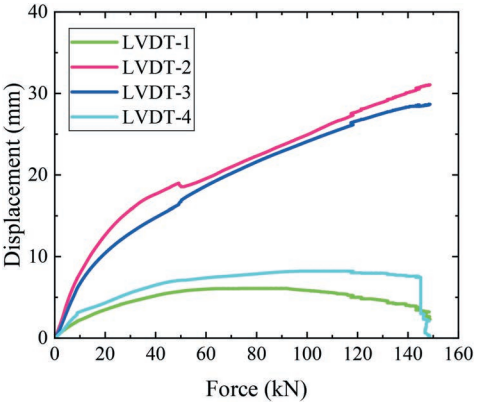


Figure 12. Force-displacement curves obtained from the loading test.

while the measurements of LVDT-1 and -4 located at the longer side started decreasing after initial increases. The sudden drop of the measurement by LVDT-4 (see the curve in cyan colour in Figure 12) represents the failure of wooden nails, which caused the top timber boards moving upwards. This demonstrates that for the prototype the load-bearing capacity of wooden nail connection is more critical than the compressive capacity of salvaged timber perpendicular to the grain, as its structural integrity (reciprocity) loses after the nail's failure.

4 DISCUSSIONS AND SUMMARY

The primary motive for exploring this interlocking system is a consequence of examining the relationship between structure and architecture using salvaged timber. In this research, the selection of form was inspired by material characteristics of salvaged timber and connection type. With a consistent cross-section but varying short lengths, salvaged timber was used in a way that only low material processing was allowed. It was this limitation that shaped the core design concept of the presented cases, which focused on utilising materials in their original states and highlighting their inherent randomness in design. From this concept, the interlocking modular system was derived, where structure and architecture were integrated: structurally, it offers considerably high structural resistance through efficiently using wooden nails, i.e., taking advantage of their tensile capacity (Ruan et al. 2022b); and architecturally, it creates a sense of rhythmic aesthetics through its layout and connection details. Such integration is different from approaches, e.g., *structure as architecture* (Charleson 2014), which define a specific form synthesising structure and architecture first. Instead, it starts from material availability, and focuses on how architecture is structured.

Structuring architecture, therefore, emphasises on material-based reasoning and the way to derive architectural form from material availability with clear structural logic. In the presented cases, the design process began with a close examination of salvaged timber, seeking not only

its structural potential but also its economic and aesthetic values. Several aspects were considered: nails were used as the only fasteners to minimise fabrication costs; despite the superior structural performance of salvaged timber when loaded parallel to the grain, it was arranged to allow flexible configurations with minimal cutting and efficient nail utilisation; and the natural irregularities and imperfections of the salvaged timber were accentuated to showcase its unique aesthetic qualities.

Structuring architecture is also an approach to pursuing sustainability. In the presented cases, multiple aspects were integrated into the design process to create a holistic solution that extends the material's lifespan and minimises material processing and waste. Beyond that, this approach also emphasises a sustainability-driven mindset for design of structures: it views a structure not merely as a "reserve" of predetermined forms or a product of calculative thinking (Heidegger 1977), but as a craft, and an art (Billington 1985) that opens up innovative possibilities to enrich architecture, making it more enduring and meaningful.

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